BUNGALOW CONSTRUCTION PLAN & COST ESTIMATION

Plot R5-180 Architectural Design, PMRDA Setback Compliance & Water Logistics | Misty Greens Phase 3B

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1. Plot Specifications & PMRDA Setback Compliance (UDCPR Maharashtra)

100% BYELAW COMPLIANT

The architectural plan for **Plot R5-180** is strictly governed by the *Unified Development Control and Promotion Regulations (UDCPR)* administered by the Pune Metropolitan Region Development Authority (PMRDA) for plots between 150 sq.m. and 250 sq.m. fronting a 9.0-meter road.

Plot Dimensions & Geometry	
Sanctioned Plot Area	**179.54 sq.m. (1,932.55 sq.ft.)**
Road Frontage (Width)	10.00 meters (32.81 ft) — Facing West
Plot Depth (Length)	17.954 meters (58.90 ft) — East-West axis
Road Width	9.0-meter wide internal paved street
Permissible Ground Coverage	Max 60% of plot area (107.72 sq.m.)

Mandatory PMRDA Setbacks & Plinth	
Front Margin (West - Road)	**3.00 meters (9.84 ft)**
Rear Margin (East - Valley)	**2.25 meters (7.38 ft)**
Side Margin (North - R5-179)	**1.50 meters (4.92 ft)**
Side Margin (South - R5-181)	**1.50 meters (4.92 ft)**
Max Plinth Footprint	**7.00m × 12.70m = 88.93 sq.m. (957.2 sq.ft.)**
Actual Ground Coverage	**49.5%** (Compliant; well under 60%)

> **Setback Audit Result:** The buildable ground footprint of **7.00m width × 12.70m depth (22.97 ft × 41.67 ft)** allows an expansive ground plinth of **957.2 sq.ft.**, yielding a total G+1 bungalow built-up area of **~2,000 sq.ft.** (FSI 1.05 to 1.10), leaving generous margins for private perimeter gardens, parking, and natural light buffers.

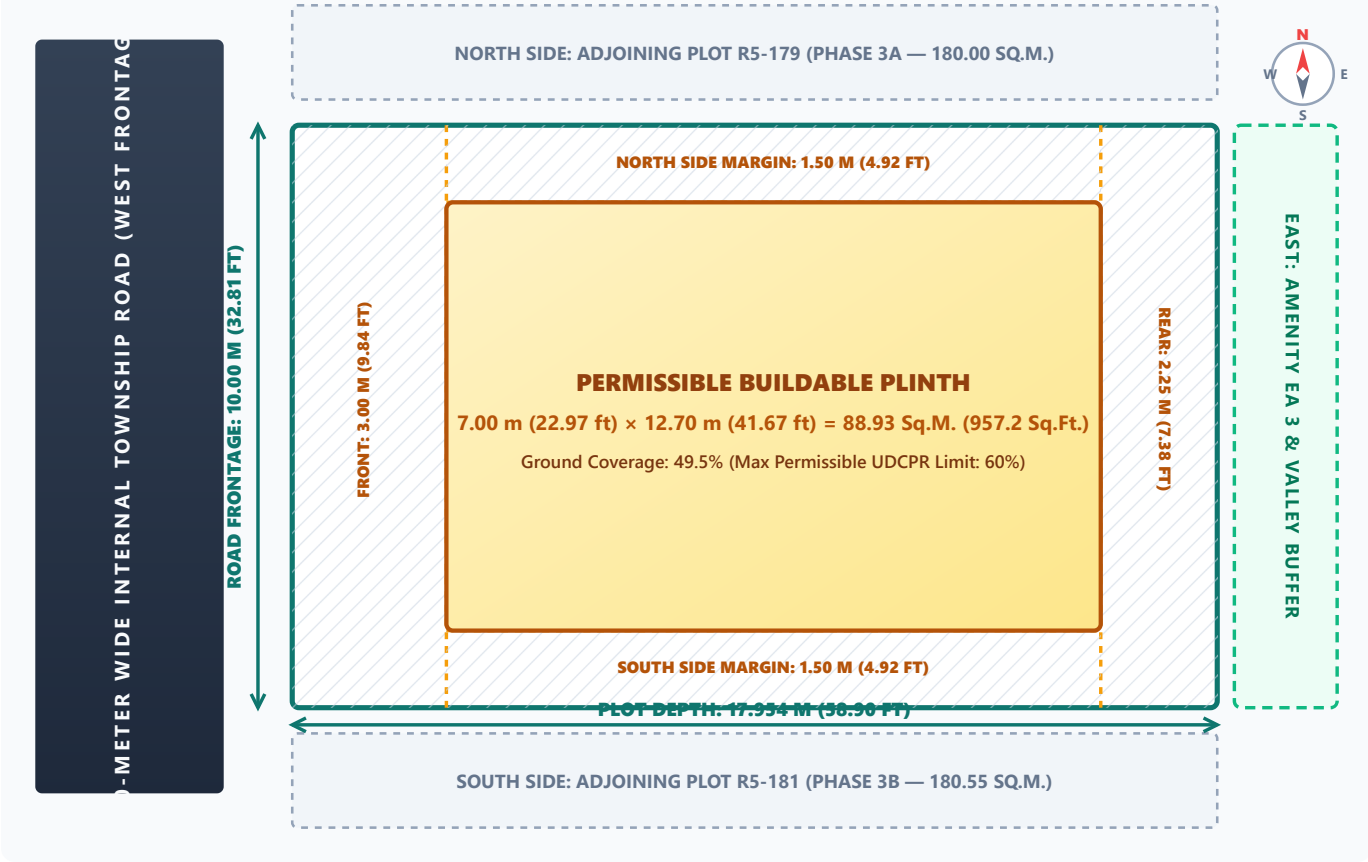


FIGURE 1.1: PMRDA SANCTIONED SETBACK BLUEPRINT & BUILDABLE PLINTH — PLOT R5-180

2. 100% Vastu Shastra Cardinal Grid & Planetary Alignment

WEST FACING MASTER HARMONIZATION

Because Plot R5-180 has road access on the **West** and backs onto the descending natural valley buffer on the **East**, the architectural spatial distribution perfectly leverages classical Vedic Vastu Shastra principles:

Zone / Pada	Deity / Cosmic Ruler	Assigned Architectural Function	Vastu Compliance & Energetic Benefit
West (Pada 4 & 3)	**Pushpadanta & Varuna**	**Main Entrance Door & Entry Foyer**	**Pushpadanta is the supreme auspicious pada on the West for wealth, learning, and progeny. Avoids Asura/Pitru padas.**
North-West (Vayu)	**Vayu Devta** (Air Element)	Covered Car Parking & Powder Room	Vayu zone governs motion and circulation. Ideal for vehicular movement and guest powder bathroom.
South-East (Agni)	**Agni Devta (Fire Element)**	**Modular Kitchen & Utility Yard**	**100% Vedic alignment. Cooking counter faces East so the cook receives beneficial morning solar ultraviolet rays.**
South-West (Nairutya)	**Nirriti / Prithvi (Earth Element)**	**Primary Master Suite (G & 1st Floor)**	**Heaviest zone with maximum structural mass. Anchors household stability, health, and family authority. Sleeping head to South.**
North-East (Ishanya)	**Shiva / Ishanya (Water Element)**	**Puja Room & Open Lawn/Water Body**	**Lightest and purest quadrant. Completely column-free with marble mandir to receive early morning cosmic prana.**
Center (Brahmasthan)	**Lord Brahma** (Space Element)	Double-Height Living Hall & Atrium	Kept completely open, light, and unencumbered. Enhances vertical airflow and natural skylight penetration.
East (Indra / Surya)	**Surya Devta** (Solar Radiance)	Dining Room & First Floor Sunrise Deck	Direct eastern exposure onto PMRDA Amenity EA 3 ensures permanent, unobstructed sunrise vistas and health vitality.
South-West (Upper)	**Saturn / Heavy Mass**	Overhead Water Tank (Terrace)	Highest structural point of the building placed in the SW, balancing the earth-gravity center.

3. Ground Floor Architectural Plan (Plinth: 957.2 Sq.Ft. | Carpet: ~820 Sq.Ft.)

SPACIOUS 3-BHK + LOUNGE CONFIGURATION

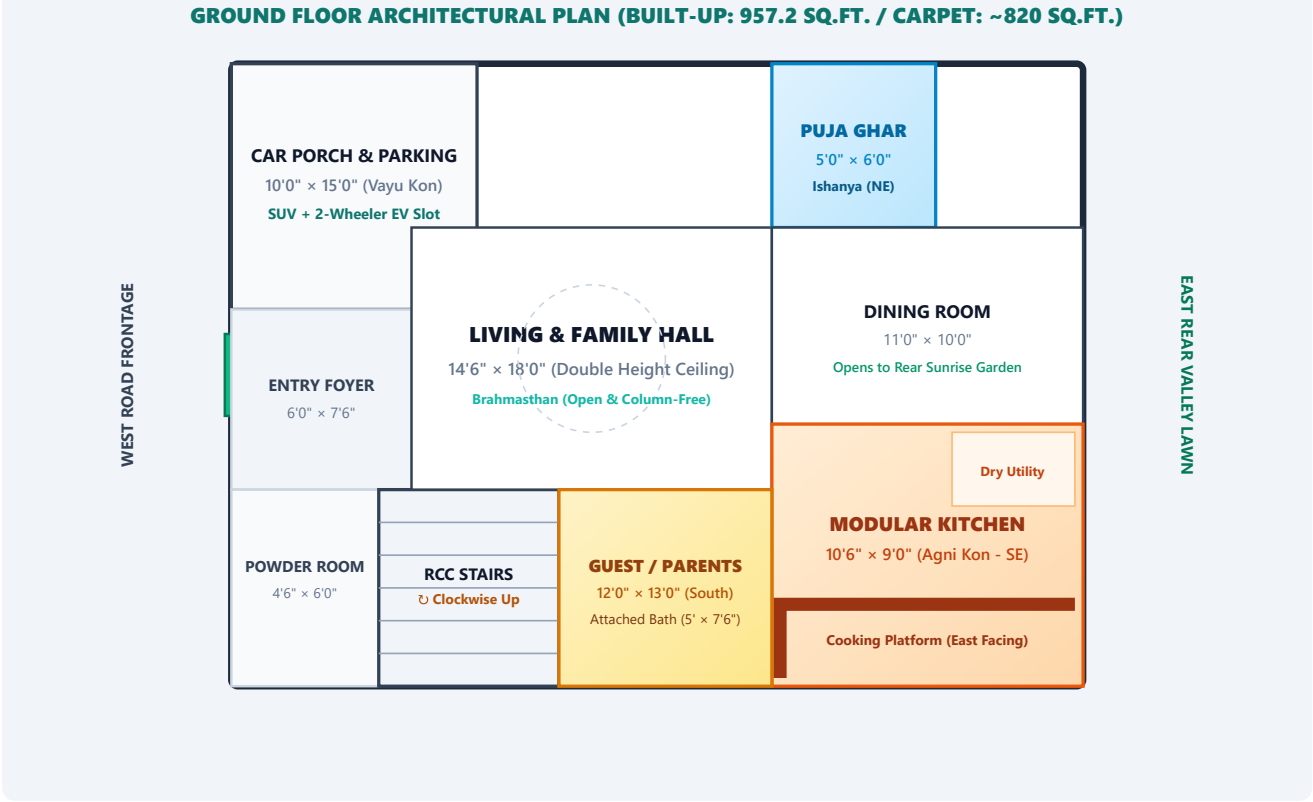


FIGURE 2.1: 2D GROUND FLOOR ARCHITECTURAL CAD LAYOUT — PLOT R5-180

Ground Floor Spatial Allocation:

- **Car Porch (NW):** 10'0" × 15'0" (Fits 1 large SUV + 2 EVs with charging).
- **Entry Foyer (West):** 6'0" × 7'6" with decorative jali screen.
- **Living & Lounge (Brahmasthan):** 14'6" × 18'0" double-height hall.
- **Dining Space (East):** 11'0" × 10'0" overlooking rear garden.
- **Modular Kitchen (SE - Agni):** 10'6" × 9'0" + 4'6" × 7'0" dry utility.
- **Puja Room (NE - Ishanya):** 5'0" × 6'0" pure marble sanctuary.
- **Guest Bedroom (South):** 12'0" × 13'0" with attached bath (5' × 7'6").
- **Powder Room (NW):** 4'6" × 6'0" for living area visitors.
- **RCC Staircase (SW):** 6'6" wide clockwise dog-legged flight.

4. First Floor Architectural Plan (Built-Up: ~1,050 Sq.Ft. with Sunrise Deck)

LUXURY MASTER SUITE & VALLEY TERRACE

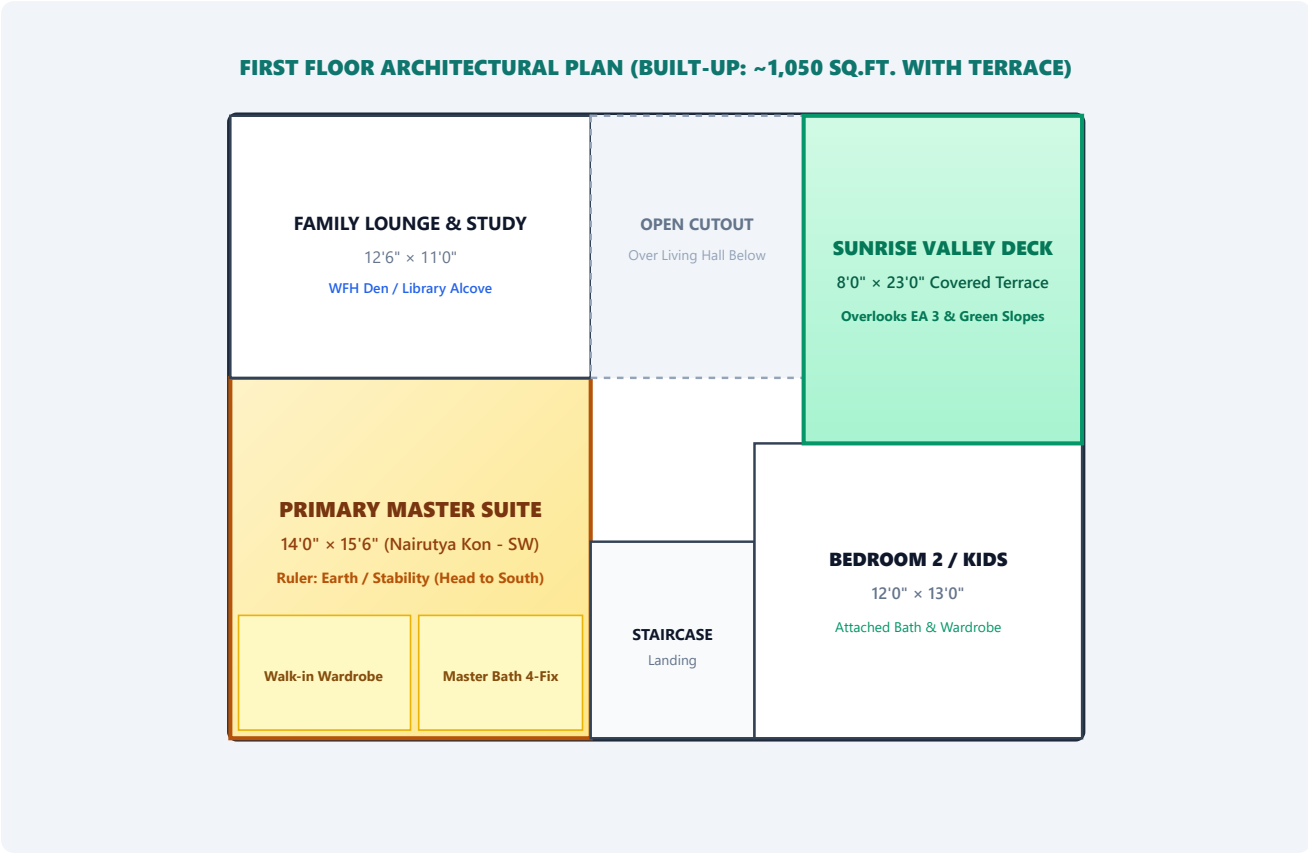


FIGURE 2.2: 2D FIRST FLOOR ARCHITECTURAL CAD LAYOUT — PLOT R5-180

First Floor Spatial Allocation:

- **Primary Master Suite (SW - Nairutya):** 14'0" × 15'6" luxury bedroom with king bed oriented to South.
- **Walk-in Dressing Closet:** 6'0" × 8'0" private dressing lobby.
- **Master En-Suite Bath:** 6'6" × 9'0" 4-fixture bath with glass cubicle.
- **Bedroom 2 / Kids Bed (SE/S):** 12'0" × 13'0" with attached bath (5' × 7'6").
- **Family Lounge & WFH Den (North):** 12'6" × 11'0" open study space.
- **Sunrise Valley Deck (East):** 8'0" × 23'0" covered cantilevered terrace framing unblocked valley sun over Amenity EA 3.

5. Hilly Terrain Engineering & Sub-Structure Specifics (Bhugaon Basalt Formation)

DECCAN BASALT STRATA

Plot R5-180 sits in the Western Ghats Deccan volcanic trap at an elevation of ~610 meters above MSL. The site exhibits hard basalt rock underlying 0.5m of murrum topsoil, with a gentle 1.2-meter natural slope from West (road level) descending toward East (Amenity EA 3 buffer).

1. Pneumatic Breaker Rock Excavation:
Standard JCB bucket excavation will refuse at 0.6m depth. Foundation requires tractor-mounted pneumatic rock breakers (chiseling) to anchor 12 isolated column footings into compact amygdaloidal basalt at 1.8m depth.

2. Plinth Leveling & UCR Retaining Wall:
To create a flat, flood-proof ground plinth (+0.6m above road crown), an Uncoursed Rubble (UCR) stone masonry retaining wall (1.2m to 1.5m height) is mandated along the

3. Anti-Termite & Damp Proofing Course (DPC):
Because the plot borders natural forested green buffers, heavy-duty chlorpyrifos chemical soil treatment is required under the plinth, plus an elastomeric polymer waterproofing coat on the exterior plinth beam faces.

4. Earthquake & Wind Resistance:
Designed for Pune Seismic Zone III (IS 1893:2016) using M25 grade concrete and Fe550D TMT bars with ductile

> eastern boundary, backfilled with gravel and boulder weep holes.

> detailing to withstand hilly gust wind pressures (basic wind speed 39 m/s).

6. Water Tanker Construction Logistics & Curing Demand Model

BHUGAON WATER MODEL

Because municipal water supply is not commissioned during active private construction in rural/peri-urban Bhugaon, **all construction and curing water must be purchased via commercial water tankers**. Below is the scientific consumption model for a G+1 RCC bungalow (~2,000 sq.ft. built-up):

Construction Stage / Activity	Curing Duration	Estimated Water Demand (Liters)	Tanker Trips (5,000L Units)
1. Site Prep, PCC & Footings	7 to 10 Days	20,000 Liters	4 Tankers
2. Plinth Beam, Retaining Wall & Compaction	14 Days	25,000 Liters	5 Tankers
3. Ground Floor Columns & Brickwork	14 Days	35,000 Liters	7 Tankers
4. First Slab Casting & Ponding Curing	**21 Days Ponding**	**50,000 Liters**	**10 Tankers**
5. First Floor Columns, Walls & Terrace Slab	21 Days Ponding	55,000 Liters	11 Tankers
6. Internal Gypsum & External Double Plaster	14 to 21 Days	40,000 Liters	8 Tankers
7. Flooring, Tiling, Washing & Labor Camp	Throughout 6 Months	25,000 Liters	5 Tankers
TOTAL CONSTRUCTION WATER DEMAND		**2,50,000 Liters**	**50 Tankers (5,000L each)**

Tanker Sourcing & Procurement Rates:
• Local supplier base: Paud Road / Bhugaon / Pirangut tanker network.
• Rate per 5,000-liter tanker: **₹ 1,300 to ₹ 1,600** (avg. ₹ 1,450).
• 50 Tankers × ₹ 1,450 = **₹ 72,500**.

On-Site Storage Infrastructure Mandate:
• 1 × 5,000L Temporary Masonry Curing Sump (brick/mortar pit): ₹ 18,000.
• 1 × 2,000L Overhead Sintex buffer tank + 0.5 HP booster pump: ₹ 14,000.
• **Total Water Budget: ₹ 1,04,500 (~₹ 1.05 Lakh)**.

7. Comprehensive Itemized Construction Cost Estimate (G+1 Villa ~2,000 Sq.Ft.)

TURNKEY PREMIUM SPECIFICATION

Turnkey construction budget for a premium, earthquake-resistant G+1 bungalow with high-grade finishes, engineered specifically for the Bhugaon micro-market:

Sr	Cost Component	Engineering Scope & Material Specifications	Estimated Cost (₹)
1	**Site Prep, Rock Excavation & Retaining**	Pneumatic breaker basalt excavation, 1.2m UCR retaining wall on East, murrum backfilling, leveling	₹ 3,10,000
2	**Construction Water Tankers & Sump**	50 × 5,000L water tankers @ ₹ 1,450 + temporary curing masonry sump + pump system	₹ 1,05,000
3	**RCC Structural Core (M25 + Fe550D)**	Footings, plinth beam, columns, beams, staircase, 2 slabs (RMC/site mix, steel ~8.5 MT @ ₹ 68/kg)	₹ 17,40,000
4	**Masonry & Plastering**	6" AAC blocks / fly ash bricks, sand-face double plaster outside, gypsum finish inside	₹ 6,60,000
5	**Waterproofing & Chemical DPC**	Terrace brickbat coba + elastomeric coat, sunken toilets, retaining wall damp proofing	₹ 1,95,000

Sr	Cost Component	Engineering Scope & Material Specifications	Estimated Cost (₹)
6	**Plumbing, Sanitation & Rainwater**	Astral/Ashirvad CPVC piping, drainage chambers, septic tank / soak pit, rainwater recharge pit	₹ 3,80,000
7	**Electrical Wiring & EV Provision**	Polycab FR copper concealed wiring, Schneider/Legrand modular plates, DBs, 15A EV charger port	₹ 3,40,000
8	**Flooring, Tiling & Granite Works**	4' × 2' GVT vitrified tiles (Kajaria/Simpolo), granite kitchen platform, anti-skid terrace/bath tiles	₹ 6,20,000
9	**Doors, Windows & Glass Railings**	Teakwood main door frame & designer shutter, UPVC sliding windows with mosquito mesh, SS/glass railings	₹ 5,10,000
10	**Painting & Exterior Elevation**	Asian Paints Royale interior emulsion, Apex Ultima weather-proof exterior silicone paint, louvers	₹ 3,50,000
11	**Compound Wall, Gate & Pavers**	6-ft perimeter compound wall, motorized MS sliding gate, heavy-duty paver blocks in front margin	₹ 3,10,000
12	**Architectural & Structural Fees**	Detailed architectural drawings, 3D elevation, RCC structural design, periodic site visits	₹ 1,80,000
13	**PMRDA Sanction & Scrutiny Fees**	PMRDA building permission fee, development charges, labor cess, sanction drawings approval	₹ 2,10,000
TOTAL ESTIMATED CONSTRUCTION BUDGET (PREMIUM TURNKEY)			**₹ 59,10,000**

Budget by Construction Quality Tier	
1. Standard Spec (Vitrified 2x2, standard UPVC)	₹ 48 Lakh – ₹ 52 Lakh (~₹ 2,450/sq.ft.)
2. Premium Turnkey (Recommended Baseline)	**₹ 56 Lakh – ₹ 62 Lakh (~₹ 2,950/sq.ft.)**
3. Luxury Villa (Automation, Solar, VRV AC)	₹ 72 Lakh – ₹ 80 Lakh (~₹ 3,800/sq.ft.)

Total Project Acquisition & Build Outlay	
Plot Land Package (Inclusive of Taxes)	₹ 1,15,00,000 (₹ 1.15 Cr)
Premium Turnkey Construction (G+1)	₹ 59,10,000 (~₹ 59.1 Lakh)
TOTAL COMBINED ASSET VALUE	**₹ 1,74,10,000 (~₹ 1.74 Cr)**